

Vinayak Thukral

Senior Software Developer · Backend & Full Stack Engineering

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SKILLS

Backend: Node.js, TypeScript, Express.js, REST APIs, GraphQL, RabbitMQ, JWT, RBAC

Frontend: React.js, Next.js, Redux, Tailwind CSS, React Native, PWA

DevOps & Cloud: Docker, CI/CD, AWS (Lambda, Amplify, IoT Core, EC2, S3), NGINX, DigitalOcean, Turborepo

Databases & ORMs: PostgreSQL, MongoDB, Firebase, Prisma ORM, Redis, SQL

EXPERIENCE

Software Developer | **Sisgain** · Noida, UP

Jan 2026 – Present

- Architected and owned scalable microservices handling high-volume API traffic, reducing average response time by 30% by designing Node.js, TypeScript, and Express.js services with PostgreSQL and Redis caching layers.
- Improved asynchronous job throughput by 45% as measured by background queue processing speed, by integrating RabbitMQ for distributed service communication and decoupled processing pipelines.
- Reduced deployment failures by 40% as measured by CI/CD pipeline success rate, by containerizing applications with Docker and enforcing Git workflow standards across the team.
- Elevated code quality across 3 junior developers by conducting structured code reviews, architecture discussions, and clean code workshops aligned with production engineering best practices.
- Strengthened API security for enterprise clients by implementing JWT-based authentication and role-based access control (RBAC), reducing unauthorized access incidents to zero post-deployment.

Software Developer | **Virtueaze** · Ahmedabad, Gujarat

Oct 2024 – Dec 2025

- Delivered a full-scale CMS platform serving 10+ agency clients, reducing client onboarding time by 35% by architecting a microservices backend using Node.js, Express.js, GraphQL, and MongoDB with Prisma ORM.
- Accelerated frontend delivery by 40% as measured by sprint velocity, by designing pixel-perfect React.js and Next.js interfaces directly from Figma prototypes with dynamic asset management modules.
- Cut infrastructure costs by 25% as measured by monthly cloud spend, by containerizing services with Docker on DigitalOcean and configuring automated NGINX deployments with backup pipelines.
- Integrated third-party payment gateways and analytics platforms into client SaaS products, increasing platform capability coverage by 60% without requiring custom builds.
- Contributed to system design discussions and database schema planning for a healthcare IoT platform, improving data query efficiency by 30% through optimized schema architecture and indexing strategies.

Full Stack Developer | **IIT Ropar** · Punjab

Feb 2024 – Sept 2024

- Boosted biomass supply chain data handling speed by 40% as measured by API response benchmarks, by engineering a TypeScript and Node.js backend with MongoDB and optimized REST API design.
- Improved frontend performance by 35% and reduced infrastructure costs by 20% as measured by AWS billing and Lighthouse scores, by containerizing with Docker and deploying via AWS Amplify with Next.js and Redux.

- Achieved real-time sensor data reliability for live IoT deployments by integrating AWS IoT Core and building a Progressive Web App, supporting offline-first field usage.
- Secured authentication for 500+ research platform users by implementing Firebase Authentication with role-based access control (RBAC), achieving zero reported auth breaches across the platform.

PROJECTS

FarmFlow · Smart Farming & IoT Platform · farmflow.ca

Stack: Node.js, React.js, AWS IoT Core, Firebase Functions, Microservices, Predictive AI

- Aggregated live sensor, weather, and soil data across regions into a unified smart farming platform, improving real-time visibility for agricultural operations at scale.
- Integrated AWS IoT Core and Firebase Functions for real-time edge processing, reducing sensor-to-dashboard latency and enabling sub-second data updates across distributed farm nodes.
- Delivered a scalable microservice architecture with full-stack React and Node.js deployment, ensuring independent scalability of data ingestion, AI inference, and user-facing services.

WebbWe · Custom CMS for Creative Agencies · webbwe.com

Stack: React.js, Node.js, Express.js, MongoDB, GraphQL, Microservices, Docker, DigitalOcean, NGINX

- Built a full-scale CMS platform enabling agencies to manage client websites, digital assets, and team roles with granular role-based access control (RBAC) and analytics dashboards.
- Designed pixel-perfect UI from Figma prototypes, implementing dynamic modules for asset uploads, templates, and client feedback workflows, reducing agency onboarding time significantly.
- Architected the backend as microservices using Node.js and Express.js, with independent services handling media management, authentication, and site publishing for fault isolation and scalability.
- Deployed services via Docker containers on DigitalOcean with automated backups and custom NGINX reverse-proxy configs, achieving high availability for production agency workloads.

MaxHealth · Healthcare & FinTech Patient Portal · portal.maxhealth.ae

Stack: Node.js, React.js, RabbitMQ, AWS (Lambda, S3), Docker, PostgreSQL, JWT, REST APIs

- Engineered a HIPAA-aligned healthcare and financial services portal handling patient records, billing workflows, and appointment management across multiple clinical and finance departments.
- Integrated RabbitMQ for asynchronous event-driven communication between healthcare microservices, ensuring reliable message delivery for critical patient data and payment processing pipelines.
- Containerized all services with Docker and deployed on AWS, leveraging Lambda for serverless event processing and S3 for secure medical document storage with role-based access control.
- Built secure REST APIs with JWT authentication and RBAC, ensuring data isolation between patient, clinician, and finance roles in compliance with healthcare data governance requirements.

EDUCATION

B.Tech, Computer Science Engineering

2025

Amritsar Group of Colleges, Punjab, India · CGPA: 7.8

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services